

DEBARGHA CHOWDHURY

COMPUTER SCIENCE UNDERGRADUATE — IoT SPECIALIZATION

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SUMMARY

Computer Science undergraduate (IoT Specialization) with backend development experience and a working foundation in software testing. Comfortable with JavaScript/TypeScript and REST APIs, with practical familiarity in unit, integration, and end-to-end testing using Mocha, Chai, and Postman. Quick learner with strong problem-solving fundamentals and genuine curiosity about hardware-software systems; eager to contribute to and grow within a QA-focused team.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java, Python, C

Testing & QA: Unit & Integration Testing, Mocha, Chai, Sinon, Postman, Basics of Mocking & Test Isolation

Backend: Node.js, Express.js, FastAPI, REST API Design, JWT Authentication, MVC Architecture

Databases & Tools: MongoDB, MySQL, Git, GitHub, GitHub Actions, Docker, Linux CLI, Vercel

IoT: Raspberry Pi, ESP32, RS485 Communication, Sensor Integration

Soft Skills: Problem-Solving, Adaptability, Team Collaboration, Technical Communication

PROJECTS

Full-Stack URL Shortener & Link Management Platform — *Node.js, Express, MongoDB, Mocha, Chai, Postman*

- Built a full-stack URL shortener with JWT authentication, MVC architecture, AI-generated link metadata (Gemini API), and MongoDB storage; deployed live on Vercel
- Added unit tests (Mocha, Chai) for slug generation, expiry logic, and auth middleware; used mocking to isolate database calls during test runs
- Built a Postman collection covering core routes (create, redirect, delete link) with checks for status code and response body, and manually explored edge/failure cases (invalid slugs, expired links)

Crop Yield Prediction System — *Raspberry Pi, Python, FastAPI, ESP32*

- Built an IoT system on Raspberry Pi with ESP32 sensors collecting soil and humidity data via RS485; FastAPI backend recommends suitable crops from collected data
- Diagnosed and fixed a sensor communication failure caused by a baud rate mismatch (9600), then added a regression check to catch the same class of issue before deployment
- Manually tested sensor-to-backend data flow across varying soil and moisture conditions to validate accuracy under real-world conditions

CERTIFICATIONS & ACHIEVEMENTS

- **Workshop on LLMs and Generative AI** — WB DSTBT-Sponsored Industrial Training Program (Feb 2025)
- **Salesforce Essentials** — Industrial Training & Internship (Aug – Oct 2025)
- **Node.js & Express.js Essentials** — Score: 87/100, NSDIP (under NITI Aayog, Govt. of India) (Jan – Apr 2025)
- **Hackfest 2025** — Led a 3-member team, coordinating problem ideation and prototype delivery under time constraints

EDUCATION

B.Tech — Computer Science & Engineering (IoT Specialization)

2023 — 2027

Meghnad Saha Institute of Technology, Kolkata, West Bengal

- Relevant Coursework: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networking, Software Engineering

Higher Secondary & Secondary Education

2021 — 2023

Burdwan CNS High School, Burdwan, West Bengal — Class XII (2023), Class X (2021)